Lab 02 - System Request

Project: Library Management System (LMS)
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1. Project Sponsor

Eng. Eman Abdel Raoof, Instructor of Systems Analysis.

2. Business Need

Currently, managing paper-based and digital documents across various departments relies heavily on manual processing, email attachments, and unorganized cloud This manual workflow leads to significant delays in document approval cycles, high risk of document loss, unauthorized access, and version confusion
Additionally, there is a lack of accountability due to the absence of a centralized audit trail
The organization needs a web-based automated Document Management System (DMS) to centralize document storage, streamline multi-level approval workflows, and enforce data security with complete audit logging.

3. Business Requirements

1. **Secure Document Upload & Organization**: Support for multiple file formats with automatic categorization, metadata tagging, and access permission settings.
2. **Automated Approval Workflows:** Configurable multi-level routing of documents to designated authority figures (Admins, Managers, Department Heads) for approval, rejection, or feedback.
3. **Audit Trail & Activity Logging:** Real-time logging of all user activities (upload, view, edit, approve, reject, delete) with timestamps and user IDs to ensure full traceability.
4. **Search & Version Control:** Advanced full-text search with metadata filters, and maintaining document version history to avoid overwriting critical data.
5. **User Management & Roles:** Admin can manage users, roles, and granular department permissions.
6. **Notifications**: System sends automated alerts for pending approvals and status updates.
7. **Reports & Export:** Department heads can view activity reports and export audit logs.
8. **Security**: Strong authentication, authorization, and data encryption.

4. Business Value

Tangible: reduces document processing and approval cycle time by over 60%; decreases paper, printing, and physical storage costs.
Intangible: enhanced data security and regulatory compliance; increased operational transparency and accountability; improved employee productivity and smoother inter-departmental collaboration

5. Special Issues and Constraints

• **Time**: limited by the course schedule.
• **Budget**: low; use open-source tools and free hosting where possible.
• **Technology**: web-based, works on desktop and mobile browsers
• **Risks**: staff resistance to adoption, migration errors of legacy digital files, data privacy compliance.
• **Out of scope**: public cloud integration, third-party electronic signature hardware.

6. Converting Requirements into Project Tasks

Business requirement	Project task (Trello card)	Lab
Need and value are clear	Write System Request	02
Is the project worth it	Feasibility Study	03-04
Choose how to work	Choose methodology	03
Who does what	Staff plan and structure	05
Plan the work	WBS and timeline	06
Requirements 1-8	Requirements specification	07
Current vs proposed workflow	DFD As-Is and To-Be	08
System behavior	UML diagrams	09
Screens (login, search, borrow)	UI mockups	10
Deliver everything	Final documentation	11

7. Project Board

Trello board: https://trello.com/invite/b/6ac7af7f3e9c3f65af5c13f8/ATTI10d6e06498620144829808d1c4dbb462CCB9D800/library-management-system

Screenshot:

